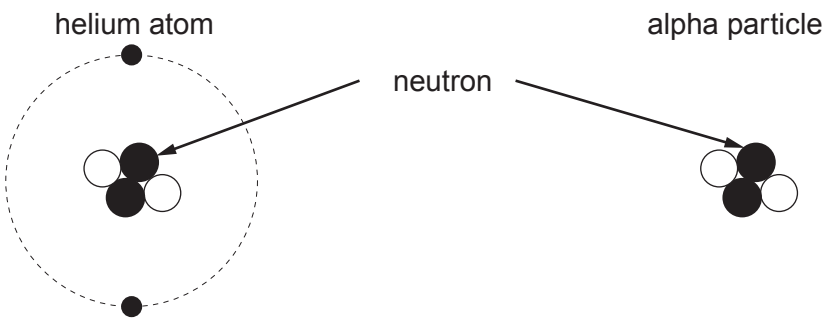


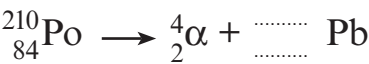
3. A student correctly draws diagrams of a helium atom and an alpha particle.



(i) One similarity between the structure of the helium atom and the alpha particle has been given in the table below. State **one** other similarity and **one** difference between them. [2]

Similarities	Difference
They both have 2 neutrons.	
.....	
.....	

(ii) A nucleus of polonium-210 decays by alpha particle emission to form lead (Pb), which is a stable element, according to the following reaction.



Complete the nuclear decay equation above. [2]

(iii) State why this isotope of polonium is radioactive. [1]

.....
.....

(iv) Explain why, in 2006, the Russian Alexander Litvinenko died within weeks of swallowing some polonium-210. [2]

.....
.....
.....

